Speaker-Aware Dialogue Act Classification

PROJECT PROPOSAL

Team name: TurnTakers

Team members: Sai Madhukar Vanam, Ramya Sruthi Pedakolimi

Chosen Task: Shared Task 1: Speaker-aware Dialogue Act Classification

Description:

We aim to explore and improve the task of Dialogue Act (DA) Classification by incorporating speaker awareness, focusing on how turns between speakers affect the meaning and intention of utterances. This is crucial for making dialogue systems more context-aware and human-like.

Our baseline is based on the paper Speaker Turn Modeling for Dialogue Act Classification (EMNLP 2021) and the associated GitHub repository, which introduces simple, yet effective speaker turn embeddings fused with utterance embeddings and processed via a Bi-GRU.

The datasets include:

- SwDA (Switchboard) – 43 DA classes, dyadic
- MRDA – 5 DA classes, multi-party
- DyDA – 4 DA classes, dyadic

Research Interest:

We are especially interested in experimenting with how well will the LLM perform on this?

Proposed Models:

i. Baseline Method:

- Speaker Turn Embedding + RoBERTa-based Utterance Embedding + Bi-GRU + Classifier (as in the original paper)

ii. Models We Hope to Try:

1. BERT with Enhanced Speaker Turn Fusion
2. CRF on Top of BERT or Bi-GRU
3. LLM-Based Few-Shot Classification

Timeline:

Date	Task	Lead Member
Apr 5–10	Set up baseline code and datasets; reproduce results	Both
Apr 11–15	Fine-tune BERT with basic speaker turn fusion	Madhukar
Apr 11–15	Add CRF layer on top of Bi-GRU for sequence modeling	Sruthi
Apr 16–20	Start LLM experiments (prompt-based few-shot GPT/T5 setup)	Sruthi
Apr 16–20	Evaluate CRF and fusion method results	Madhukar
Apr 25–May 5	Final exams – no project work scheduled	-
May 6–7	Final model comparison, performance analysis, visualization	Both
May 8	Submission	